Comparative Evaluation of Classical, Deep Learning, and Quantum-Enhanced Architecture For Lung Cancer Classification from CT Images

Haritha Ramadass
School of Computing
Dublin City University
Dublin, Ireland
haritha.ramadass2@mail.dcu.ie

Abstract*— Worldwide, lung cancer is the main cause of cancer-related deaths. Computed tomography (CT) continues to serve as the primary modality for early detection. Despite extensive machine learning studies in this domain, controlled comparisons of classical, deep, transformer-based, and quantum-enhanced architectures under identical experimental conditions are lacking in the literature, precluding evidence-based model selection for clinical deployment. This paper presents such a comparison across four learning paradigms for three-class lung cancer classification (Normal, Benign, Malignant) on the IQ-OTH/NCCD CT dataset: a custom Convolutional Neural Network (CNN), a Support Vector Machine (SVM) with ResNet-50 deep feature extraction, a fine-tuned Vision Transformer (VIT), and a hybrid Quantum Vision Transformer (QVIT). All four models were trained and evaluated on an identical balanced dataset of 360 CT images (120 per class, downsampled from 1,190 original images) using a fixed, stratified 70/30 split and four-fold stratified cross-validation, with accuracy, macro-averaged precision, recall, F1-score, and ROC-AUC as evaluation criteria. The CNN achieved the highest test-set performance (accuracy 93.52%, macro F1 0.9353), followed by SVM (92.59%, macro F1 0.9256) and VIT (92.59%, macro F1 0.9254), with QVIT lowest (89.81%, macro F1 0.8952). Grad-CAM, SHAP, and CLS-to-patch attention visualisation confirmed anatomically plausible evidence of decision-making among all four models. CNNs perform better due to their ability to learn local image features and textures effectively, especially with limited data. QVIT performance is constrained by current quantum hardware, where the 768-dimensional ViT features must be compressed into 12 inputs, causing information loss before quantum processing. This limitation is due to hardware constraints rather than the hybrid quantum-classical approach itself. SVM with deep transfer features acts as a strong, computationally effective classical alternative. These results constitute the first controlled four-paradigm benchmark on this dataset and provide reproducible baselines for future hybrid quantum medical imaging research.

Index Terms*— Lung cancer classification, computed tomography imaging, convolutional neural network, support vector machine, vision transformer, hybrid vision transformer.

1. Introduction

Globally, lung cancer is the leading cause of cancer-related death. It causes around 1.8 million deaths annually, or nearly 18% of all cancer-related deaths [1], [2]. Despite ongoing clinical advancements, five-year survival rates remain below 20%, primarily due to late-stage detection, at which point curative interventions are limited [2], [3]. Computed tomography (CT) is the established modality for lung cancer screening, enabling high-resolution detection of subcentimeter nodules. Nevertheless, accurate CT interpretation requires specialised radiological expertise, is time-intensive, and is affected by inter-examiner variability. These problems have driven persistent interest in machine learning-based computer-aided diagnosis [3], [4].

Lung cancer CT classification has progressed through three distinct generations of machine learning. Early methods combined handcrafted radiomic features with Support Vector Machines (SVMs) and kernel techniques, forming foundational benchmarks [4], [5], [7]. Subsequently, Convolutional Neural Networks (CNNs) outperformed earlier methods by learning hierarchical spatial representations from data and removing the requirement for manual feature engineering [8], [11], [12]. Vision Transformers (ViTs), introduced by Dosovitskiy et al. [16], replaced local convolution with global self-attention across the image patches, achieving strong performance on large-scale medical imaging tasks [18], [19], [20]. More recently, hybrid quantum-classical architectures, enabled by differentiable quantum emulation frameworks such as PennyLane [24], have appeared as a new paradigm, providing unique representational capabilities within the constraints of current Noisy Intermediate-Scale Quantum (NISQ) devices [22], [23], [25].

A considerable methodological gap persists within the existing literature: each generation of models has typically been validated in isolation, employing different datasets, category distributions, preprocessing methods, and evaluation protocols. As a result, reported performance differences often reflect experimental design choices rather than intrinsic architectural capabilities, making the selection of models for clinical application more complex and less reliable. To date, no published study has simultaneously evaluated CNN, SVM, VIT, and a hybrid Quantum Vision Transformer (QVIT) under a unified protocol using the same balance. This study directly handles this methodological-related gap. We conduct a controlled comparative evaluation of four distinct learning paradigms for three-class lung cancer CT classification—Normal, Benign, and Malignant—using the IQ-OTH/NCCD dataset [27]. All models are trained and evaluated on an identical balanced dataset (360 images, 120 per class), with a fixed stratified 70/30 split (seed = 42), four-fold stratified cross-validation, a unified preprocessing pipeline, and a consistent multi-metric evaluation framework including Grad-CAM [28], SHAP [29], and attention visualisation. By holding all experimental variables constant, we isolate architectural properties as the source of observed performance differences, establishing the first reproducible four-paradigm benchmark on this dataset .

This study is guided by two research questions. RQ1: Under a controlled and reproducible experimental protocol, which architecture—CNN, SVM, VIT, or hybrid QVIT—achieves the highest three-class classification performance on the IQ-OTH/NCCD CT dataset, and can the QVIT perform competitively with established classical and deep learning standards? RQ2: How do the feature-learning mechanisms and interpretive ability characteristics of CNN, SVM, VIT, and QVIT differ, and what practical trade-offs do these differences present for clinical deployment?

2. Related Work

Lung cancer CT classification has advanced through successive generations of machine learning, each introducing distinct assumptions about how to extract discriminative structure from imaging data. This section reviews classical, deep learning, transformer-based, and quantum-enhanced approaches, and then identifies the methodological gap that motivates the present comparative study.

2.1. Classical Machine Learning for Lung CT Classification

Classical approaches relied on handcrafted feature extraction combined with discriminative classifiers. Radiomic descriptors—texture statistics, shape measurements, and intensity histograms from segmented nodule regions—were paired with SVMs, k-nearest neighbours, and random forests to build early automated diagnostic systems [4], [5]. Vapnik's maximum-margin theory [5] made SVMs particularly useful in medical imaging: margin maximisation provides strong generalisation in high-dimensional spaces, and the kernel formulation permits non-linear boundaries without explicit feature transformations.

On the IQ-OTH/NCCD dataset, Kareem et al. [7] applied SVMs (linear, RBF, polynomial) to features from bit-plane slicing, Gabor filtering, and GLCM analysis, achieving 89.9% accuracy without class balancing — the most directly comparable classical baseline for this study. However, handcrafted pipelines have a fundamental limitation: descriptor design requires domain expertise, offers limited flexibility, and may not capture the full discriminative structure of CT pathology. Contemporary SVM pipelines address this by replacing handcrafted features with deep transfer-learned representations, improving expressiveness while retaining margin-based classification guarantees [4], [6].

2.2. CNN-Based Deep Learning Approaches

CNNs transformed lung cancer classification by eliminating the need for manual feature engineering through end-to-end hierarchical spatial learning [12]. Convolutional filters learn local texture/shape detectors in early layers and abstract these into semantic representations in deeper layers — well suited to the localised textural signatures of pulmonary nodules (typically 5–30 mm) [3], [8]. Weight sharing across spatial positions reduces the number of parameters with fully connected networks, providing an inductive bias that matches the spatial stationarity of nodule appearances across slices [12].

On IQ-OTH/NCCD, Musthafa et al. [9] used a double-layered CNN with SMOTE oversampling and Gaussian blur, reporting 99.64% accuracy; however, SMOTE's pixel-space interpolation produces synthetic images that don't reflect authentic pathology, raising generalisability concerns. Abe and Nyathi [10] used a novel Random Pixel Swap (RPS) augmentation, achieving 97.56% accuracy and 98.61% AUC-ROC across CNN and transformer architectures, and confirmed horizontal flipping as an effective, anatomically conservative augmentation. Haziq et al. [11] reported further gains through architectural enhancements and parameter optimisation. Notably, none of these studies evaluates CNNs against SVM, VIT, and quantum baselines under identical conditions — the comparison this study provides.

2.3. Vision Transformers in Medical Imaging

Dosovitskiy et al. [16] introduced the Vision Transformer, showing that sequences of linearly embedded patches processed through multi-head self-attention can match or exceed CNN performance when pretrained on sufficiently large datasets. Unlike CNNs, which expand receptive fields gradually with depth, VITs compute pairwise attention across all patch tokens from the first layer, enabling broad contextual reasoning CNNs must otherwise approximate through successive convolutions — clinically relevant for thoracic CT, where bilateral lung symmetry, mediastinal relationships, and diffuse parenchymal patterns are global features CNNs capture inefficiently at shallow depths.

Shamshad et al. [18] and Azad et al. [19] provide comprehensive reviews showing that VITs consistently approach CNN performance across segmentation, classification, and diagnosis tasks, with advantages most evident when using large pretraining datasets and transfer learning. Pal et al. [20] demonstrated competitive VIT performance for lung and colon cancer classification. Raghu et al. [21] found VITs develop qualitatively different internal features than CNNs, attending globally from early layers rather than building local-to-global hierarchies, suggesting complementary rather than strictly superior capabilities. A consistent finding is that VITs' advantage over CNNs is not assured on small datasets, where weaker spatial inductive bias can be disadvantageous — examined directly here using a fold-level training set of 189 images.

2.4. Quantum and Hybrid Quantum-Classical Learning

Quantum Machine Learning (QML) combines quantum computing with machine learning to improve how data is represented and processed. Parameterised quantum circuits can encode and transform data in ways that are difficult for classical methods to achieve, potentially improving feature representation, kernel approximation, and optimisation [22]. Biamonte et al. [22] established the foundations of QML and identified hybrid quantum-classical models as the most practical approach for Noisy Intermediate-Scale Quantum (NISQ) devices, which are limited by a small number of qubits and short coherence times. Havlíček et al. [23] further showed that quantum feature maps can produce decision boundaries that are difficult for classical kernel methods to replicate.

This study uses PennyLane [24], which enables hybrid quantum-classical models to be trained using the parameter-shift rule within standard deep learning frameworks. In medical imaging, Martis et al. [25] achieved 92.12% accuracy on chest CT and radiograph classification using a hybrid quantum deep learning model, making it the closest published benchmark to the QVIT model evaluated in this research.

Despite these promising results, current NISQ hardware has important limitations. Medical images must be compressed before being encoded into quantum circuits, creating an information bottleneck that has not been systematically studied for medical image classification. In addition, McClean et al.

[26] identified the barren plateau problem, where gradients become exponentially small in deeper quantum circuits, making them difficult to train. These challenges motivated the use of a shallow two-layer, 12-qubit QVIT architecture in this study.

3. Dataset and Experimental Setup

This section describes the dataset, data preprocessing steps, and evaluation methodology supporting the controlled comparison of four paradigms. All experimental design decisions were held constant across CNN, SVM, VIT, and QVIT to ensure that the observed performance differences in Section 5 are attributable to architectural properties rather than to variations in data preparation or evaluation protocols.

3.1. Dataset Description and Class Balancing

The IQ-OTH/NCCD Lung Cancer Dataset [27] contains 1,190 thoracic CT images from 110 patients at the Iraqi Oncology Teaching Hospital and the National Centre for Cancer Diseases. Images were acquired using Siemens SOMATOM scanners at 1 mm axial slice thickness and are organised into three diagnostic categories: Normal, Benign, and Malignant. The current dataset exhibits substantial class imbalance, with malignant and normal slices considerably outnumbering benign cases. While this distribution reflects clinical prevalence, it creates a risk of training bias that can inflate accuracy on majority classes at the expense of sensitivity for smaller classes.

Class balance was achieved through stratified random downsampling to 120 images per class, resulting in a balanced dataset of 360 images. Downsampling was selected instead of synthetic oversampling techniques such as SMOTE, as SMOTE generates new samples by interpolating between existing instances in pixel space, producing CT image blends that do not correspond to real pathological appearances [9]. Training classifiers on such simulated samples may inflate reported test accuracy beyond what would generalise to clinically unseen data. Downsampling makes sure that every training sample is a genuine CT acquisition, preserving the authentic image distribution at the cost of a smaller effective training set. This trade-off prioritises experimental validity over maximum reported performance.

3.2. Preprocessing Pipeline

A unified preprocessing pipeline was applied consistently across all four models to isolate architectural differences from preprocessing effects. All models utilised a single input resolution of 224×224 pixels. For VIT and QVIT, this resolution is a strict architectural requirement: the google/vit-base-patch16-224-in21k checkpoint divides images into 16×16 patches, and its positional embedding matrix is pretrained for exactly 196 tokens at this resolution. Any alternative input size would result in an incompatible token count, thereby invalidating the pretrained weights [16]. Although CNN and SVM do not share this constraint, the 224×224 resolution was maintained for pipeline consistency and to maximise spatial detail without increasing the sizes of fully connected layers or risking overfitting on the 252 training images.

All images were normalised using the ImageNet channel-wise mean and standard deviation ($\mu = [0.485, 0.456, 0.406]$, $\sigma = [0.229, 0.224, 0.225]$). This normalisation is required for the pretrained weights of VIT and QVIT and was applied to CNN and SVM inputs for consistency. Random horizontal flipping ($p = 0.5$) was the only augmentation employed, as it is the sole geometric transform that preserves anatomical plausibility in axial thoracic CT slices. Vertical flipping inverts diaphragm-to-apex orientation, and large rotations generate implausible parenchymal geometry. This conservative approach is consistent with Abe and Nyathi [10], who found horizontal flipping effective on this dataset, even when combined with more aggressive augmentations

For SVM, no augmentation was applied during ResNet-50 feature extraction because the model operates on a fixed 2048-dimensional feature vector per image. Stochastic augmentation would yield inconsistent and irreproducible representations, potentially destabilising the linear decision boundary. Only evaluation-mode transforms (resize and normalise) were used, ensuring that each image maps to a single deterministic feature vector.

3.3. Data Partitioning and Cross-Validation

The balanced dataset of 360 images was partitioned using a stratified 70/30 split with a fixed random seed (seed = 42), resulting in 252 training images (84 per class) and 108 test images (36 per class). The test set was held out entirely until final model evaluation and did not participate in any aspect of model development. The test set was excluded from training, validation, and model parameter selection. Stratified four-fold cross-validation was applied within the training set for model development and stability assessment. With 252 training images across four folds, each fold used 189 images for training (63 per class) and 63 images for validation (21 per class). Full reproducibility was ensured by applying a fixed random seed (seed = 42) to dataset partitioning, model weight initialisation, and batch ordering, together with the cuDNN deterministic computation mode.

3.4. Evaluation Protocol

Macro-averaged metrics were designated as the primary outcome because they assign equal importance to all three diagnostic classes regardless of support, which is essential given that the minority group (Benign) has clinical importance equal to that of the majority classes. One-vs-rest ROC AUC was computed per class to assess probabilistic discrimination independently of the classification threshold. Cross-validation stability was reported as mean ± standard deviation across four folds, providing evidence that test-set rankings are not artefacts of a single partition.

4. Model Architectures And Design

Each of the four models—CNN, SVM, VIT, and QVIT—was designed to represent a distinct learning paradigm rather than incremental variations on a single approach. This strategy ensures that the comparison in Section 5 reflects genuine architectural variation rather than superficial configuration differences. All four models utilise a unified preprocessing pipeline and a 224×224 input resolution, as described in Section 3, and differ only in their approach to processing the input for classification. This section presents the design, training configuration, and rationale for each model.

4.1. Convolutional Neural Network (CNN)

The CNN serves as the classical deep learning baseline in this comparison, learning hierarchical spatial properties directly from CT images without handcrafted preprocessing or a pretrained backbone. The design and training configuration are summarised below.

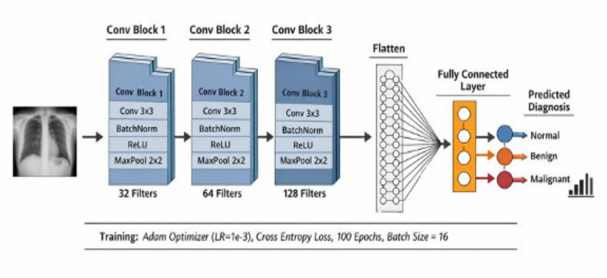

Fig 1.Convolutional Neural Network Model

The CNN (Fig. 1) consists of three sequential convolutional blocks, each with a 3×3 convolutional layer, batch normalization [13], ReLU activation, and 2×2 max pooling. Filter count increases from 32 to 64 to 128 channels, following the hierarchical abstraction convention established in the deep learning literature [12]. An input resolution of 224×224 was selected to maximize spatial detail. The architecture requires no fixed input size, given the absence of pretrained-weight constraints, and this resolution is used uniformly across all four models in the study. Three pooling operations reduce the input to a 28×28 spatial map, flattened to 128×28×28 = 100,352 features, then passed through fully connected layers FC(100,352→256) and FC(256→3). This provides sufficient resolution to represent the 5–30 mm texture characteristics of pulmonary nodules while remaining computationally practical for a 252-image training set. Batch normalization after each convolutional layer reduces internal covariate shift and provides implicit regularisation [13], particularly useful for the compact 256-unit fully connected layer in limiting overfitting-prone parameters at this dataset scale.

With a learning rate of 1×10^{-3}, which is typical for training convolutional architectures from random initialization, the network was trained using the Adam optimizer [15]. Adam's adaptive per-parameter scaling accommodates differing gradient scales across early texture-detecting layers and deeper semantic layers, removing the need for manual learning-rate scheduling [15]—a property validated for CNN-based lung cancer CT classification [8]. This learning rate produces gradient steps large enough for randomly initialized weights to break symmetry and converge within a practical training budget, consistent with established guidance for scratch-trained convolutional networks [15]. Training used a batch size of 16, yielding 16 gradient updates per epoch (252 ÷ 16). This configuration has been shown to balance gradient stability and update frequency for CNN-based classification on small medical CT datasets [8], [11]. Over 100 epochs, this totals 1,600 gradient updates, sufficient for full convergence on a 252-image training set without overfitting, consistent with epoch budgets reported in comparable lung CT CNN studies [8]. CrossEntropyLoss—the standard and theoretically ideal objective for balanced multi-class image classification—was used as the training loss. It directly maximizes the predicted probability of the correct class by combining log-softmax activation with negative log-likelihood [12].

4.2. Support Vector Machine with Deep Feature Extraction

The SVM illustrates the classical machine learning paradigm in this comparison. However, it differs from the handcrafted-feature approach discussed in Section 2.1 by integrating a linear support vector classifier with deep transfer-learned features rather than manually engineered descriptors.

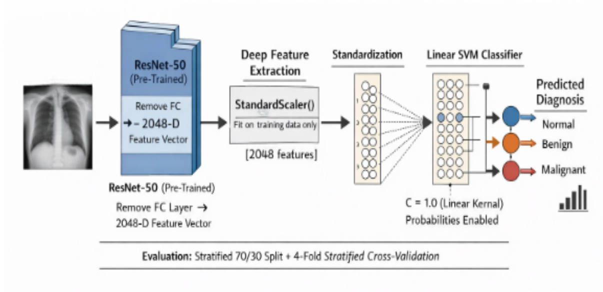

Fig 2. SVM Pipeline with ResNet-50 Feature Extraction

The SVM pipeline (Fig. 2) separates representation learning from decision-boundary optimization. A pretrained ResNet-50 [14] backbone, with its final classification layer removed, serves as a fixed feature extractor. Each CT image passes through 50 layers of convolutional, batch-normalized, and residual-connected transformations, producing a 2048-dimensional global average pooling vector. Pretraining on 1.4 million ImageNet images enables the backbone to capture texture and structural visual features without domain-specific feature engineering [14], which suits the limited training set size. Rather than training a deep network end-to-end on 252 images—likely causing severe overfitting—the pretrained ResNet-50 representations are used directly as fixed, semantically rich classifier inputs. This is reflected in the performance gain: ResNet-50 features raise classification effectiveness from 89.9% with handcrafted features [7] to 92.59%, suggesting improved representational quality.

A linear kernel was chosen for the support vector classifier because ResNet-50's convolutional, batch-normalized, and residual transformations project CT images into a feature space where diagnostic classes are approximately linearly separable [4], [5]. A linear decision boundary is thus both theoretically appropriate and computationally efficient, consistent with established practice for deep transfer-learned features [4]. The regularisation parameter was set to the standard default C = 1.0 [6], which balances the margin for these inputs. This choice aligns with ResNet-50's training regime, which incorporates weight decay and batch normalization [6].

Probability estimation used Platt scaling, fitting a sigmoid calibration curve to the SVM's decision function outputs to generate well-calibrated probabilities. This calibration supports one-vs-rest ROC-AUC curves and SHAP-based interpretability outputs, both requiring continuous probability estimates rather than discrete class labels [6], [19]. The 2048-dimensional feature vectors were standardized using a StandardScaler fitted exclusively to the training partition and applied to the test partition without refitting, preventing information leakage. Standardizing features to zero mean and unit variance is standard preprocessing for linear SVCs, ensuring each ResNet-50 dimension contributes proportionally to the margin calculation rather than being skewed by scale differences [6]. This pipeline—combining deep transfer-learned features, standardization, and a linear decision boundary—aligns with methodologies validated in SVM-based medical image classification studies [4].

4.3. Vision Transformer (VIT)

The Vision Transformer (VIT) incorporates the transformer paradigm into this comparison, substituting the convolutional neural network's local convolutional inductive bias and the support vector machine's fixed deep features with fine-tuned global self-attention applied to image patches.

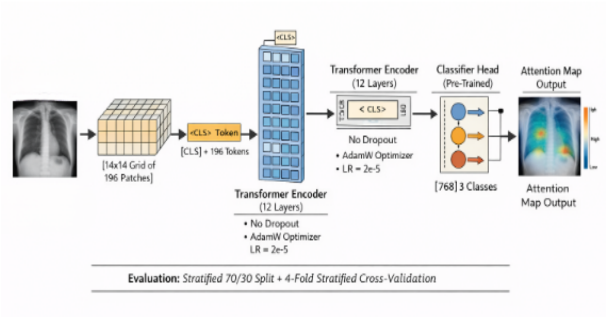

Fig 3. Architecture of the Vision Transformer (VIT) Model

Fine-tuning uses the google/vit-base-patch16-224-in21k checkpoint [16], pretrained on ImageNet-21k (14M images, 21,843 classes). The 224×224 input matches the checkpoint's positional embedding matrix (196 patch positions), partitioning each image into 196 non-overlapping 16×16 patches, linearly embedded with learnable positional encodings. Each patch covers ~16 mm of tissue at standard CT field of view—sufficient to capture clinically significant nodule features while keeping self-attention tractable [16]. A prepended CLS token aggregates global context across 12 transformer layers (12 attention heads each), enabling the model to jointly attend to nodule location, surrounding tissue, and bilateral lung structure [16]. Self-attention computes pairwise relationships among all 197 tokens from the first layer, supporting global contextual reasoning relevant to bilateral and mediastinal patterns [18], [21], beyond the localised receptive fields of CNNs. The final 768-dimensional CLS embedding feeds a three-class classifier head.

ViT-Base (86M parameters, 12 layers), pretrained on ImageNet-21k, was chosen for its broad, transferable visual prior across 21,843 categories [16]. Attention and hidden-state dropout were set to zero, letting pretrained weights provide the primary regularisation, supplemented by a fine-tuning learning rate low enough to preserve pretrained attention structure.

Fine-tuning used AdamW [17], which decouples weight decay from adaptive gradient scaling—the recommended strategy for VIT fine-tuning [16] and a fix for a known inconsistency in standard Adam [17]—at a learning rate of 2×10^{-5}, validated for fine-tuning ViTs on small datasets [16]. Batch size was 8, the maximum supported for full-precision 197-token forward passes under available GPU memory, consistent with standard ViT fine-tuning configurations [16]. Training ran 100 epochs (~3,100 updates), matching the CNN's epoch budget for a fair, epoch-controlled comparison.

4.4. Hybrid Quantum Vision Transformer (QVIT)

The QVIT serves as the quantum-enhanced paradigm in this comparison, evaluating whether a variational quantum circuit can extend the pretrained VIT representation into a novel computational substrate — directly addressing RQ1 by testing whether such an architecture can compete with established classical and deep learning baselines.

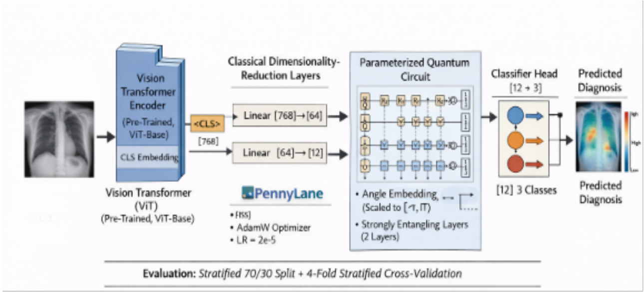

Fig 4. Architecture of the Hybrid Quantum Vision Transformer

The QVIT (Fig. 4) combines the pretrained VIT backbone with a parameterised quantum circuit in a hybrid quantum-classical transfer-learning pipeline [22], [24]. The ViT encoder's 768-dimensional CLS embedding is reduced via a two-stage classical bottleneck: Linear(768→64) with GELU, then Linear(64→12). The intermediate 64-dimensional layer enables non-linear feature selection before the final 12-dimensional reduction; GELU maintains consistency with the ViT backbone and reflects the hierarchical dimensionality-reduction principle of deep residual architectures [14].

The 12-dimensional latent vector is scaled by $\tanh(x) \times \pi$ into $(-\pi, \pi]$, giving full Bloch-sphere coverage, smooth differentiable behaviour, no boundary gradient saturation, and preserved feature-magnitude ordering—valid inputs for rotation gates. It is encoded into a 12-qubit circuit via AngleEmbedding [24] (one gate per qubit, constant depth), consistent with standard angle-based encoding [24]. Twelve qubits match the compressed latent dimension, with four qubits per diagnostic class, aligning the quantum state space with the three-class task.

StronglyEntanglingLayers [24] is applied across the twelve qubits, each layer using Euler-decomposed single-qubit rotations (Rz·Ry·Rz, three parameters/qubit) with cyclic CNOT entanglement—72 trainable quantum parameters total—accessing the full per-qubit Hilbert space with strong expressivity for the three-class boundary while keeping circuit depth suited to stable gradient-based optimisation on the available data [26]. Pauli-Z expectation values from each qubit yield 12 real outputs, passed to a final linear layer mapping to three diagnostic classes.

All quantum operations ran on PennyLane's default qubit simulator [24], computing exact analytic expectation values for a reproducible, noise-free evaluation of the quantum layer's contribution. The full QVIT was trained end-to-end with AdamW [17] at 2×10^{-5}, matching the VIT backbone's fine-tuning configuration. Quantum parameters were jointly optimised via the parameter-shift rule through PennyLane's TorchLayer interface [24], enabling coherent classical-quantum co-training, with AdamW's decoupled weight decay preserving the pretrained ImageNet-21k representations throughout.

5. Results

This section presents the comparative evaluation results across all four models, structured from aggregated to per-class performance, followed by cross-validation stability and ROC-based probabilistic discrimination. Positioning against prior published work on the same dataset closes the section.

5.1. Aggregated Performance

Table 1 presents test-set performance measures for all four models evaluated on the held-out 108-image test set. The CNN demonstrated the strongest results across all primary metrics, achieving 93.52% accuracy and a macro F1 Score of 0.9353. Both the SVM and VIT produced nearly identical aggregated results, each attaining 92.59% accuracy, with macro F1-scores of 0.9256 and 0.9254, respectively.

TABLE 1. Aggregated Test-Set Performance Metrics (n=108)

Metric	CNN	SVM	ViT	QViT
Accuracy	**0.9352**	0.9259	0.9259	0.8981
Macro Precision	**0.9357**	0.9269	0.9258	0.9014
Macro Recall	**0.9352**	0.9259	0.9259	0.8981
Macro F1-Score	**0.9353**	0.9256	0.9254	0.8952
Weighted Precision	**0.9357**	0.9269	0.9258	0.9014
Weighted Recall	**0.9352**	0.9259	0.9259	0.8981
Weighted F1-Score	**0.9353**	0.9256	0.9254	0.8952

These outcomes are notable given the distinct learning mechanisms: the SVM utilises fixed deep features with a linear decision boundary, while the VIT employs end-to-end fine-tuned global self-attention. The QVIT lagged behind the three standard models, achieving 89.81% accuracy and a macro F1 of 0.8952, representing a 3.71 percentage-point deficit relative to the CNN. The alignment of accuracy and macro F1 rankings across all four models indicates that the performance ordering is not an artefact of class weight selection. The near-identical macro and weighted F1-scores within each model confirm that the balanced class distribution (36 images per class in the test set) yields equivalent macro and weighted outcomes. This property of perfectly balanced test partitions validates the downsampling strategy. The CNN's advantage stays consistent across all seven metrics, eliminating the possibility that its lead is due to single-metric optimisation artefacts.

5.2. Per-Class Diagnostic Analysis

Per-class analysis (Table 2) reveals clinically significant distinctions that aggregated metrics do not capture. The CNN's overall lead is primarily attributable to its performance in the Malignant class, achieving a precision of 0.97 and an F1 score of 0.96. This represents the strongest malignant-class result among all models and is the most clinically critical outcome, as malignant false negatives entail the highest risk of delayed treatment. Additionally, the CNN achieves the highest Benign recall (0.94) among all models, indicating balanced sensitivity across all three diagnostic categories rather than specialisation to a single class.

The VIT achieves the highest Normal recall (0.94), indicating that global self-attention is especially effective at characterising structurally normal lung tissue. In this context, the absence of localised pathology across the entire scan serves as the discriminative signal, making global context more informative than local texture detection. The QVIT's performance deficit is concentrated and class-specific rather than uniformly distributed: Benign recall (0.85) and Malignant recall (0.87) are both substantially lower than those of the standard models (0.92–0.94), while Malignant precision (0.92) remains comparatively close to the SVM and VIT. This suggests that the 72-parameter quantum circuit produces conservative class boundaries, systematically missing positive cases rather than generating false positives. Two findings from Table 2 are directly relevant to clinical deployment. First, no model yields a Malignant recall below 0.87, indicating that all four models detect the majority of malignant cases, which is the primary clinical safety requirement.

TABLE 2. Per-Class Test-Set Metrics (n=108, 36 per class)

Class	Metric	CNN	SVM	ViT	QViT
Normal	Precision	0.92	0.93	0.92	0.89
	Recall	0.92	0.93	**0.94**	0.87
	F1-Score	0.92	0.93	0.93	0.88
Benign	Precision	0.92	0.92	0.93	0.89
	Recall	**0.94**	0.92	0.92	0.85
	F1-Score	**0.93**	0.92	0.92	0.87
Malignant	Precision	**0.97**	0.94	0.93	0.92
	Recall	**0.94**	**0.94**	0.93	0.87
	F1-Score	**0.96**	0.94	0.93	0.89

Second, the QV*I*T's Benign recall of 0.85 implies that 15% of Benign cases are misclassified. At this dataset scale, this corresponds to approximately 5–6 images per test fold being incorrectly assigned, predominantly to the Normal or Malignant classes. For a clinical support tool, this level of Benign misclassification would call for human radiologist review of all borderline predictions rather than autonomous diagnostic decision-making.

5.3. Cross-Validation Stability

Cross-validation results support and extend the test-set rankings. The CNN achieves the highest mean macro F1 (0.93) and mean AUCROC (0.97), but also exhibits the largest fold-to-fold variance (accuracy SD = 0.06, F1 SD = 0.05). This variance reflects the model's sensitivity to the specific 189 of 252 images included in each training fold when learning from random initialisation. The SVM and VIT demonstrate the lowest variance among all models (accuracy and F1 SD = 0.01), which is consistent with their reliance on stable pretrained representations: the linear SVC utilises fixed ResNet50 features, and the VIT fine-tunes from a well-established ImageNet-21k prior, rather than learning representations from scratch. The QVIT's moderate variance (accuracy SD = 0.02, AUC-ROC SD = 0.03) is intermediate between the scratch-trained CNN and the transfer-learned models, matching a circuit whose 72 quantum parameters are sensitive to training subset composition but less volatile than the CNN's full parameter set. The consistent AUC-ROC of 0.96 across SVM, VIT, and QVIT cross-validation folds demonstrates that probabilistic class discrimination is stable even where hard-label accuracy shows greater variation.

TABLE 3. Four-Fold Cross-Validation Summary (Training Set, n=252, Mean ± SD)

Model	Mean Accuracy	Mean Macro F1	Mean AUC-ROC
CNN	0.92 ± 0.06	$\mathbf{0.93 \pm 0.05}$	$\mathbf{0.97 \pm 0.02}$
SVM	$\mathbf{0.92 \pm 0.01}$	0.92 ± 0.01	0.96 ± 0.01
ViT	$\mathbf{0.92 \pm 0.01}$	0.92 ± 0.01	0.96 ± 0.01
QViT	0.90 ± 0.02	0.90 ± 0.02	0.96 ± 0.03

5.4. ROC Analysis

Fig. 5 presents per-class one-vs-rest ROC curves for all four models on the held-out test set. All four models attain an AUC above 0.94 for each class, confirming robust probabilistic class discrimination across the full range of prediction scores. This finding holds even for the QVIT, despite its hard-label accuracy trailing the standard models by 3.71 percentage points.

Three ROC findings are noteworthy. First, the CNN achieves the highest Benign AUC (0.99) among all models, consistent with its highest Benign recall in Table 2, confirming that the CNN's probabilistic score distribution for Benign cases is the most discriminative. Second, the SVM, VIT, and QVIT all achieve a perfect Malignant AUC (1.00), indicating that their probability scores rank malignant cases above non-malignant cases at every threshold, whereas the CNN achieves a Malignant AUC of 0.97. This is a nuanced result: the CNN attains a higher hard-label Malignant F1 (0.96 versus 0.94 for SVM) but a lower Malignant AUC (0.97 versus 1.00 for SVM), suggesting that the CNN makes more confident correct predictions at the default threshold, whereas the SVM provides a more continuously discriminative probability value across all thresholds.

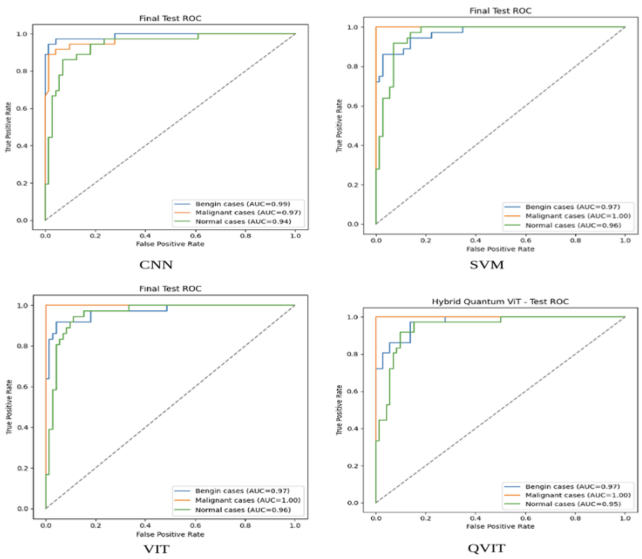

Fig 5. ROC Curve Comparison for all models

Third, the QVIT achieves a Malignant AUC of 1.00 despite its lower recall (0.87 at the default threshold), demonstrating that its quantum circuit produces well-ordered malignant probability scores even when the decision boundary at threshold 0.5 misclassifies some malignant cases. This evidence suggests that threshold recalibration, rather than architectural replacement, is the appropriate clinical intervention.

5.5. Comparison with Existing Literature

Comparison with earlier work on IQ-OTH/NCCD requires close consideration, as reported accuracies are highly dependent on balancing and augmentation approaches that vary across studies. Kareem et al. [7] reported 89.9% accuracy using a polynomial-kernel SVM with handcrafted Gabor and GLCM features without class balancing. The SVM in the present study exceeds this by 2.7 percentage points using a simpler linear kernel, thereby isolating feature encoding rather than kernel complexity as the source of improvement, given that the classifier family and dataset are held constant. Musthafa et al. [9] and Abe and Nyathi [10] show higher CNN-family accuracies (99.64% and 97.56%), but both adopt techniques deliberately avoided in this study: SMOTE-based synthetic interpolation and Random Pixel Swap augmentation on imbalanced data. As discussed in Section 3.1, synthetic interpolation can inflate accuracy by training on images that do not reflect real pathology. The CNN result reported here should therefore be interpreted as a controlled, downsampling-based baseline that prioritises generalisation reliability over peak reported accuracy. Notably, Abe and Nyathi's VIT result was not their top-performing configuration; CutMix outperformed RPS on that architecture, underscoring that single reported figures should not be taken as an architecture's ceiling without accounting for the augmentation used.

Besides individual comparisons, the principal contribution of this study is methodological. No earlier work on this dataset has evaluated CNN, SVM, VIT, and a hybrid QVIT together under a single controlled protocol, with a shared preprocessing pipeline and a balanced partition. This addresses the confound identified as the central research gap in Section 2. Previous studies have applied individual architectures to this dataset, but each used distinct splits, balancing, and augmentation approaches, rendering cross-study performance differences inseparable from experimental design. The controlled design implemented here directly eliminates that confound: the observed performance ordering (CNN > SVM ≈ VIT > QVIT) is attributable to model design, with a level of confidence that isolated studies cannot provide. This conclusion is further supported by cross-validation stability (Table 3) and the ROC discrimination reported in Section 5.4.

The class-balancing strategy adopted in this study (120 images per class), which differs from all prior work cited, likely contributes to the stable cross-validation variance observed in Table 2. This approach delivers a reusable template for controlled balancing and matched evaluation in future comparative studies on this and similar datasets.

6. Interpretability Analysis

Interpretability is essential for the clinical deployment of diagnostic AI systems; models with inexplicable predictions cannot be trusted in patient-facing workflows [29]. This section presents interpretability results for all four models, employing GradCAM [28] for the CNN, SHAP [29] for the SVM, and CLS-to-patch attention visualisation for the VIT and QVIT. A cross-model comparison of decision evidence quality is presented below.

6.1. CNN — Grad-CAM

Grad-CAM [28] was applied to the final batch normalisation layer of the CNN's third convolutional block. This method computes the gradient of the predicted class score with respect to feature map activations, generating a spatial heatmap that highlights class-discriminative regions.

Fig 6 presents Grad-CAM outputs for representative test cases. Activation is consistently concentrated in the lung parenchyma and nodule-adjacent tissue, with minimal response in thoracic wall structures, diaphragm boundaries, or background regions outside the lung field. This spatial localisation aligns with clinical expectations, as the CNN focuses on regions that a radiologist would examine when assessing a pulmonary lesion.

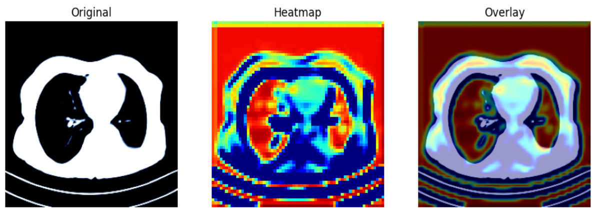

Fig 6. GRAD CAM for CNN

The absence of activation on non-anatomical artefacts such as scanner edges, table boundaries, or patient positioning markers provides evidence that the CNN's 93.52% accuracy reflects authentic learning of pathological features rather than dataset-specific shortcut correlations. The high Malignant precision (0.97, Table 2) further supports this, as confident malignant predictions are accompanied by concentrated heatmaps over lesion tissue.

6.2. SVM — SHAP Feature Attribution

SHAP KernelExplainer [29] was applied to the 2048-dimensional ResNet-50 feature vectors extracted from each test image, computing Shapley values that attribute the SVM's predicted class probability to individual feature dimensions using a model-agnostic, perturbation-based approach.

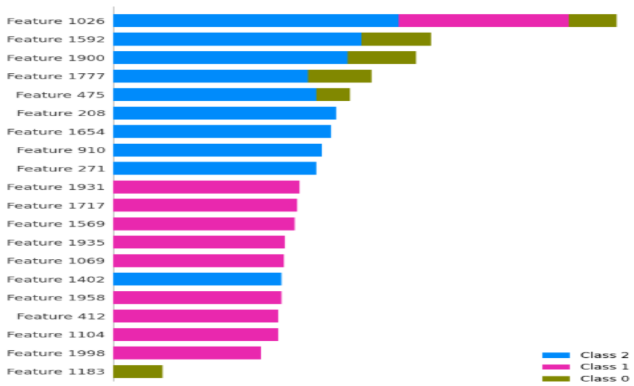

Fig 7. SHAP Values for SVM

Fig 7 demonstrates that a small subset of ResNet-50 dimensions dominate the SVM's classification boundaries. Feature 1026 exhibits the strongest overall predictive weight across all three classes. Features 1931 and 1717 primarily drive Benign-class predictions, while Features 1592 and 1900 are specific to Malignant-class predictions. This sparse, class-discriminative attribution pattern is consistent with the SVM's strong per-class quality in Table 2 and confirms that the linear decision boundary makes use of a genuinely discriminative subspace of the 2048-dimensional ResNet-50 embedding, rather than assigning weight uniformly. As SHAP analysis operates in latent feature space rather than image space, direct spatial interpretability is limited; Feature 1026 cannot be directly mapped to a specific visual structure without further analysis of ResNet-50 activations. Nevertheless, the class-specificity of attributions provides quantitative evidence that the SVM's classification is based on structured feature-class associations rather than noise.

6.3. VIT and QVIT — Attention Visualisation

CLS-to-patch attention parameters were extracted from the final transformer layer for both VIT and QVIT. These weights were upsampled to the original image resolution using bilinear interpolation and overlaid on the input CT slice to generate spatial attention maps.

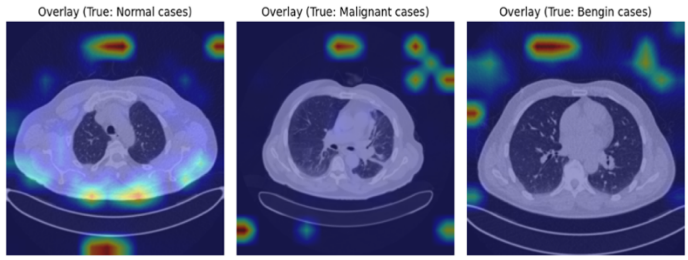

Fig 8. Attention maps for a few test images in VIT

VIT attention maps (Figure 8) display spatially coherent concentration over thoracic structures spanning all three diagnostic classes. In malignant cases, high-weight patches cluster over lesion-bearing lung regions and adjacent parenchyma.

In benign cases, attention localises to nodule boundaries and surrounding tissue. In normal cases, attention is distributed across bilateral lung fields, consistent with VIT's Normal recall advantage (0.94, Table 2), where the overall contextual signal from the bilateral normal parenchyma provides information. These focus patterns correspond to the spatial reasoning employed by trained radiologists, supporting the radiological plausibility of VIT's 92.59% accuracy.

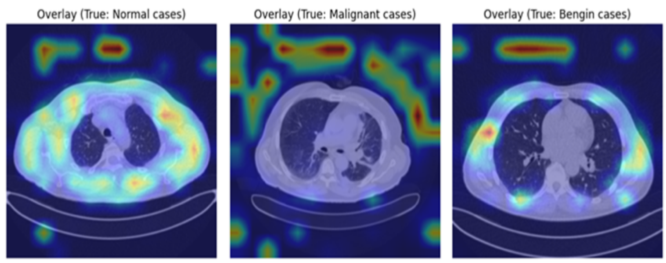

Fig 9. Attention maps for a few test images (Hybrid Quantum VIT)

QVIT attention maps (Fig 9) exhibit more diffuse, spatially distributed patterns for matched test cases, despite both models sharing the same VIT encoder up to the CLS token. High-weight patches are distributed across a wider region of the image, with lower peak intensities and reduced anatomical specificity compared to VIT maps. Since the VIT backbone is identical in both models, this difference is attributed to the information bottleneck introduced by the 768→64→GELU→12 compression and the downstream quantum circuit. This compression alters the gradient flow through the backbone during fine-tuning, reducing the spatial coherence of CLS-token attention, even though the attention computation remains unchanged. This interpretation aligns with the QVIT's reduced Benign recall (0.85) and Malignant recall (0.87) in Table 2, as diffuse attention yields less precise class boundaries for the diagnostically challenging Benign and Malignant categories.

6.4. Cross-Model Interpretability Comparison

Across all four models, no evidence was found of reliance on non-anatomical image artefacts such as scanner edges, patient positioning markers, or table boundaries. All models direct their decision evidence toward the lung parenchyma, mediastinal structures, or nodule-adjacent tissue, confirming that high performance reflects genuine pathological feature learning rather than dataset-specific shortcuts [28], [29]. In terms of spatial directness together with clinical verifiability, the ranking is as follows: CNN (Grad-CAM) ≥ VIT (attention) > QVIT (attention) > SVM (SHAP). The CNN provides the most immediately radiologist-verifiable explanations, permitting practitioners without machine learning expertise to inspect a Grad-CAM overlay and assess whether the highlighted region corresponds to the suspected lesion. The SVM offers the most quantitatively rigorous attribution via Shapley values [29], but requires additional steps to translate latent-feature attributions into spatial regions of the image. The QVIT's diffuse attention maps most clearly reveal its architectural limitation, functioning as both a clinical weakness and a mechanistic insight.

A key cross-model finding is the noted alignment between interpretability quality and classification performance in this study. The CNN, which achieves the highest accuracy, also produces the most anatomically specific and clinically verifiable explanations. This agreement is a desirable property for clinical decision-support tools, as model correctness alone is insufficient for deployment without corresponding explainability [29].

7. Discussion and Conclusion

The CNN's strong performance reflects alignment between its architectural prior and the task: convolutional weight sharing encodes the spatial locality of pulmonary nodules, reducing sample complexity and directly benefiting performance on 252 training images. Raghu et al. [19] showed CNNs exhibit stronger texture bias while ViTs exhibit stronger shape bias — an advantage here, since the Benign–Malignant distinction is primarily textural (nodule density, margin sharpness, infiltration). Grad-CAM confirms this mechanistically, with activation consistently localised to nodule boundaries and lesion-adjacent parenchyma.

The SVM's 92.59% accuracy, trailing the CNN by only 0.93 points, shows architectural complexity is unnecessary once a rich fixed representation exists: ResNet-50's 2048-dimensional embedding renders classes approximately linearly separable [4], [12], reducing the SVM's task to a maximum-margin boundary in an already favourable space. This yields the lowest cross-validation variance of any model (SD = 0.01, Table 4) and, with 15-minute training and CPU-only inference, makes the SVM the most resource-efficient option.

The ViT's failure to exceed the CNN (92.59% vs 93.52%) reflects self-attention's lack of spatial inductive bias — an advantage at scale but a liability with only 252 training images [14], [19]. Its one advantage, Normal recall (0.94 vs 0.92, Table 3), fits this pattern: recognising the *absence* of localised pathology benefits from global attention, while the harder Benign–Malignant distinction favours the CNN's local feature hierarchy at this sample size.

The QViT's 89.81% accuracy reflects three compounding constraints: the 768→64→GELU→12 compression removes over 98% of the ViT's representational capacity before quantum encoding, degrading attention coherence (Fig. 9); 72 trainable quantum parameters are insufficient to capture the full within-class distribution, producing high precision (0.92) but low recall (0.85–0.87, Table 3) [24]; and two variational layers were the maximum depth feasible before barren plateaus stall convergence [24]. These are consequences of current quantum hardware and dataset scale, not fundamental barriers to the hybrid paradigm.

Clinically, all four models exceed 89% accuracy with plausible interpretability, positioning them as decision-support tools rather than autonomous systems — even the CNN's 93.52% still misclassifies 6–7 cases per 108, and missed malignant cases carry the highest cost. The CNN's combination of the highest Malignant F1 (0.96) and the most anatomically specific Grad-CAM explanations makes it the strongest candidate for a radiologist-facing interface. The QViT's Malignant AUC of 1.00 despite lower recall suggests its deficit is a calibration issue, addressable through threshold recalibration rather than redesign.

This paper presented a controlled comparative evaluation of CNN, SVM, ViT, and hybrid QViT for three-class lung cancer classification on IQ-OTH/NCCD under identical experimental conditions, attributing performance differences to architecture rather than experimental design. **RQ1** is answered by the CNN's strongest performance across all metrics (93.52% accuracy, macro F1 0.9353), driven by its spatial inductive bias; the QViT remains a functional but not yet competitive proof of concept (89.81% accuracy, Malignant AUC = 1.00), its gap attributable to addressable NISQ-era constraints rather than a fundamental limitation of quantum computation. **RQ2** is answered by the mechanistic and interpretability analyses above: all four models use qualitatively distinct mechanisms, yet SVM and ViT converge to near-identical accuracy through entirely different pathways. For deployment, the CNN offers the strongest overall combination of accuracy and interpretability; the SVM is the most resource-efficient; the ViT demands greater computational investment for equivalent accuracy; and the QViT remains research-stage, appropriate where quantum feasibility rather than clinical performance is the priority.

The QViT's results should not be read as a ceiling for quantum machine learning in medical imaging — its bottleneck, parameter budget, and depth constraint reflect applying NISQ-era simulation to a 252-image dataset, not fundamental properties of quantum computation. As hardware scales and datasets grow, hybrid quantum-classical architectures remain a compelling direction, for which this study establishes a reproducible, controlled benchmark [20], [22].

8. Limitations

This study evaluated four architecturally distinct paradigms under a controlled, identical experimental protocol—the central contribution of the work—and the boundaries of that protocol are noted here as directions for extension rather than deficiencies in the comparison itself. The working dataset, balanced to 360 images from a single institution and scanner model, was chosen to eliminate inter-scanner variability as a confound, ensuring observed performance differences reflect architecture rather than acquisition variation; broader multi-centre validation remains a natural next step before any deployment claim [16]. Performance differences are reported as cross-validation means and standard deviations across four folds, providing a reasonable estimate of model stability;

however, formal statistical significance testing across folds—for example, a paired test on fold-level macro F1—was not performed, and the observed CNN-led ranking should therefore be interpreted as descriptive rather than statistically confirmed. The QVIT was evaluated on PennyLane's default qubit analytic simulator, which computes exact, noise-free expectation values — the methodologically correct choice for isolating the quantum circuit's representational contribution from hardware noise—but extending this evaluation to physical NISQ devices remains a valuable next stage as quantum hardware matures. Finally, the QVIT's classical and quantum parameters were optimised under a single unified learning rate to preserve training consistency with the VIT backbone; decoupled learning rates for the two parameter groups are identified as a promising refinement for future architectures.

9. Future Research Directions

Building on these findings, four directions extend the benchmarking framework established here. The most immediate is multi-centre external validation on datasets such as LUNA16 and LIDC-IDRI, to assess performance across institutions and scanner manufacturers [16]. A second direction is to expand the effective training set through anatomy-preserving augmentation—such as Random Pixel Swap [10]—or generative augmentation via conditional diffusion models—thereby growing the training pool from 252 to 500–1,000 real images without synthetic artefacts, particularly benefiting VIT and QVIT. With a larger pool, a third direction becomes viable: increasing the QVIT's circuit depth beyond two layers and testing three- to five-layer StronglyEntanglingLayers configurations to assess whether added expressibility improves classification [26]. Finally, executing the QVIT on physical NISQ hardware, such as IBM Quantum or IonQ, would characterise the impact of decoherence and gate error relative to the simulator results reported here, informing whether error mitigation is needed for clinical competitiveness.

10. Generative AI Usage Disclosure

Generative AI was used only for language editing and creating architectural diagrams. All outputs were reviewed and validated by the authors. All research, methodology, implementation, analysis, and conclusions are the authors' original work.

11. References

[1] H. Sung, J. Ferlay, R. L. Siegel, M. Laversanne, I. Soerjomataram, A. Jemal, and F. Bray, "Global cancer statistics 2020: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries," *CA: A Cancer Journal for Clinicians*, vol. 71, no. 3, pp. 209–249, 2021.

[2] F. Bray, M. Laversanne, H. Sung, J. Ferlay, R. L. Siegel, I. Soerjomataram, and A. Jemal, "Global cancer statistics 2022: GLOBOCAN estimates of incidence and mortality worldwide for 36 cancers in 185 countries," *CA: A Cancer Journal for Clinicians*, vol. 74, no. 3, pp. 229–263, 2024.

[3] Y. Li, X. Wu, P. Yang, G. Jiang, and Y. Luo, "Machine learning for lung cancer diagnosis, treatment, and prognosis," *Genomics, Proteomics & Bioinformatics*, vol. 20, no. 5, pp. 850–866, 2022.

[4] K.-L. Hua, C.-H. Hsu, S. C. Hidayati, W.-H. Cheng, and Y.-J. Chen, "Computer-aided classification of lung nodules on computed tomography images via deep learning technique," *OncoTargets and Therapy*, pp. 2015–2022, 2015.

[5] V. Vapnik, *The Nature of Statistical Learning Theory*. New York, NY, USA: Springer Science & Business Media, 2013.

[6] F. Pedregosa, G. Varoquaux, A. Gramfort, V. Michel, B. Thirion, O. Grisel, M. Blondel, P. Prettenhofer, R. Weiss, V. Dubourg, et al., "Scikit-learn: Machine learning in Python," *Journal of Machine Learning Research*, vol. 12, pp. 2825–2830, 2011.

[7] H. F. Kareem, M. S. AL-Husieny, F. Y. Mohsen, E. A. Khalil, and Z. S. Hassan, "Evaluation of SVM performance in the detection of lung cancer in marked CT scan dataset," *Indonesian Journal of Electrical Engineering and Computer Science*, vol. 21, no. 3, p. 1731, 2021.

[8] S. Nageswaran, G. Arunkumar, A. K. Bisht, S. Mewada, J. S. Kumar, M. Jawarneh, and E. Asenso, "[Retracted] Lung cancer classification and prediction using machine learning and image processing," *BioMed Research International*, vol. 2022, no. 1, p. 1755460, 2022.

[9] M. M. Musthafa, I. Manimozhi, T. R. Mahesh, and S. Guluwadi, "Optimizing double-layered convolutional neural networks for efficient lung cancer classification through hyperparameter optimization and advanced image pre-processing techniques," *BMC Medical Informatics and Decision Making*, vol. 24, no. 1, p. 142, 2024.

[10] A. A. Abe and M. Nyathi, "Lung cancer diagnosis from computed tomography images using deep learning algorithms with random pixel swap data augmentation: Algorithm development and validation study," *JMIR Bioinformatics and Biotechnology*, vol. 6, no. 1, p. e68848, 2025.

[11] U. Haziq, J. Uddin, S. Rahman, M. Yaseen, I. Khan, J. Khan, and Y. Jung, "Improving lung cancer detection with enhanced convolutional sequential networks," *Scientific Reports*, vol. 15, no. 1, p. 32099, 2025.

[12] I. Goodfellow, Y. Bengio, and A. Courville, *Deep Learning*. Cambridge, MA, USA: MIT Press, 2016.

[13] N. Bjorck, C. P. Gomes, B. Selman, and K. Q. Weinberger, "Understanding batch normalization," *Advances in Neural Information Processing Systems*, vol. 31, 2018.

[14] M. Shafiq and Z. Gu, "Deep residual learning for image recognition: A survey," *Applied Sciences*, vol. 12, no. 18, p. 8972, 2022.

[15] D. P. Kingma and J. Ba, "Adam: A method for stochastic optimization," *arXiv preprint* arXiv:1412.6980, 2014.

[16] A. Dosovitskiy, L. Beyer, A. Kolesnikov, D. Weissenborn, X. Zhai, T. Unterthiner, M. Dehghani, M. Minderer, G. Heigold, S. Gelly, et al., "An image is worth 16x16 words: Transformers for image recognition at scale," *arXiv preprint* arXiv:2010.11929, 2020.

[17] I. Loshchilov and F. Hutter, "Decoupled weight decay regularization," *arXiv preprint* arXiv:1711.05101, 2017.

[18] F. Shamshad, S. Khan, S. W. Zamir, M. H. Khan, M. Hayat, F. S. Khan, and H. Fu, "Transformers in medical imaging: A survey," *Medical Image Analysis*, vol. 88, p. 102802, 2023.

[19] R. Azad, A. Kazerouni, M. Heidari, E. K. Aghdam, A. Molaei, Y. Jia, A. Jose, R. Roy, and D. Merhof, "Advances in medical image analysis with vision transformers: A comprehensive review," *Medical Image Analysis*, vol. 91, p. 103000, 2024.

[20] A. Pal, H. M. Rai, J. Yoo, S.-R. Lee, and Y. Park, "VIT-DCNN: Vision transformer with deformable CNN model for lung and colon cancer detection," *Cancers*, vol. 17, no. 18, p. 3005, 2025.

[21] M. Raghu, T. Unterthiner, S. Kornblith, C. Zhang, and A. Dosovitskiy, "Do vision transformers see like convolutional neural networks?," *Advances in Neural Information Processing Systems*, vol. 34, pp. 12116–12128, 2021.

[22] J. Biamonte, P. Wittek, N. Pancotti, P. Rebentrost, N. Wiebe, and S. Lloyd, "Quantum machine learning," *Nature*, vol. 549, no. 7671, pp. 195–202, 2017.

[23] V. Havlíček, A. D. Córcoles, K. Temme, A. W. Harrow, A. Kandala, J. M. Chow, and J. M. Gambetta, "Supervised learning with quantum-enhanced feature spaces," *Nature*, vol. 567, no. 7747, pp. 209–212, 2019.

[24] V. Bergholm, J. Izaac, M. Schuld, C. Gogolin, S. Ahmed, V. Ajith, M. S. Alam, G. Alonso-Linaje, B. AkashNarayanan, A. Asadi, et al., "Pennylane: Automatic differentiation of hybrid quantum-classical computations," *arXiv preprint* arXiv:1811.04968, 2018.

[25] J. E. Martis, S. MS, B. R, A. M. Mutawa, and M. Murugappan, "Novel hybrid quantum architecture-based lung cancer detection using chest radiograph and computerized tomography images," *Bioengineering*, vol. 11, no. 8, p. 799, 2024.

[26] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven, "Barren plateaus in quantum neural network training landscapes," *Nature Communications*, vol. 9, no. 1, p. 4812, 2018.

[27] M. AL-Huseiny *et al.*, "The IQ-OTH/NCCD lung cancer dataset," *Mendeley Data*, vol. 4, 2023.

[28] R. R. Selvaraju, M. Cogswell, A. Das, R. Vedantam, D. Parikh, and D. Batra, "Grad-CAM: Visual explanations from deep networks via gradient-based localization," *International Journal of Computer Vision*, vol. 128, no. 2, pp. 336–359, 2020.

[29] S. M. Lundberg and S.-I. Lee, "A unified approach to interpreting model predictions," *Advances in Neural Information Processing Systems*, vol. 30, 2017.